Transportation Exposure & Environmental Justice in Massachusetts (2022)

A GIS Analysis of Traffic Proximity, Air Quality, and Demographic Disparities

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Tools: ArcGIS Pro, EJScreen 2022, US Census ACS 2022, Excel

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Project Overview

This project evaluates whether transportation exposure and air pollution burdens vary across income and minority populations in Massachusetts. While fine particulate matter ($PM_{2.5}$) appears relatively uniform statewide, transportation-related indicators — particularly the **Traffic Proximity Index (PTRAF)** — reveal localized disparities associated with highways, rail corridors, and urban centers.

The project integrates spatial analysis, environmental justice (EJ) indicators, and demographic grouping to identify patterns of disproportionate exposure. A zoomed analysis of the Greater Boston region provides additional context for dense urban transportation networks.

Research Objectives

- Measure $PM_{2.5}$ and PTRAF patterns across Massachusetts census tracts.
- Evaluate differences in transportation exposure between **Low**, **Mid**, and **High** income and minority groups.
- Identify whether transportation corridors correspond to elevated EJ burden.
- Compare uniform statewide $PM_{2.5}$ concentrations with more variable transportation exposure indicators.
- Visualize spatial patterns using statewide and inset maps.

Methods

Data Preparation

1. Downloaded **EPA EJScreen 2022** dataset for Massachusetts.
2. Clipped features to the Massachusetts state boundary.
3. Removed geometry-only or null-value tracts.
4. Calculated two categorical variables:
    - **Income_Group** (Low, Mid, High) using LOWINCPCT

- **Minority_Group** (Low, Mid, High) using MINORPCT

$PM_{2.5}$ Baseline Analysis

- Evaluated statewide $PM_{2.5}$ concentrations by income group.
- Found minimal variation (~6.78–6.82 μg/m³).
- Exported $PM_{2.5}$ summary table and map layout.

PTRAF Transportation Exposure Analysis

- Mapped PTRAF values across Massachusetts.
- Added transportation context using:
    - MBTA commuter rail (statewide)
    - MBTA rapid transit (Boston metro)
- Calculated mean PTRAF by:
    - Income Group (Low, Mid, High)
    - Minority Group (Low, Mid, High)
- Exported summary tables and created a bar chart of exposure differences.

Visualization

- Produced:
    - Statewide $PM_{2.5}$ map
    - Statewide PTRAF map with city labels + Boston inset map
    - PTRAF bar chart
    - "Why Income ≠ Minority" infographic to explain demographic gradients

All outputs were assembled into a final project PDF.

Data Sources

EPA AQS $PM_{2.5}$ Annual Summary (2022)
Fully quality-assured $PM_{2.5}$ data with minimal variance across Massachusetts.
https://aqs.epa.gov/aqsweb/airdata/download_files.html

EPA EJScreen 2022
Census-tract-level environmental justice indicators including $PM_{2.5}$, PTRAF, DSLPM, LOWINCPCT, MINORPCT.
https://www.epa.gov/ejscreen/download-ejscreen-data

US Census ACS / TIGER 2022
Tracts, demographic percentages, boundaries.
https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html

MassGIS Transportation Data
MBTA rapid transit and commuter rail systems for contextual analysis.
https://www.mass.gov/info-details/massgis-data-mbta-rapid-transit

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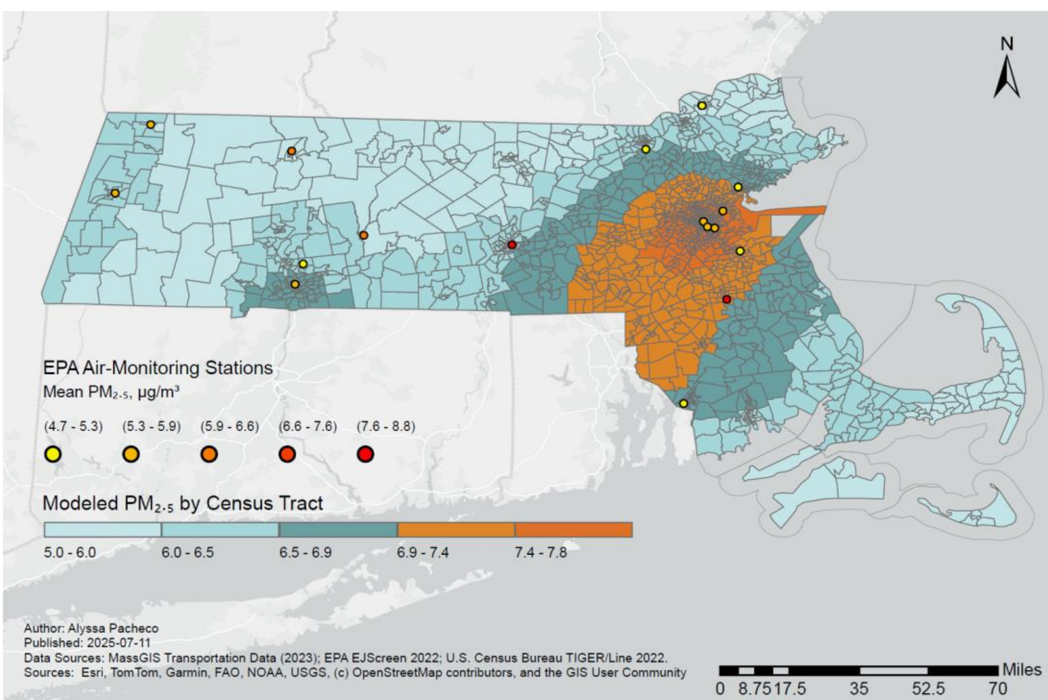
Results

$PM_{2.5}$ (Fine Particle Pollution)

- $PM_{2.5}$ was **remarkably stable** across income groups.
- Statewide values ranged only from **6.78–6.82 µg/m³**.
- Suggests no environmental justice disparity in **ambient air quality**.

Fine Particulate Pollution ($PM_{2.5}$) Across Massachusetts, 2022
Modeled concentrations by census tract with EPA monitoring sites

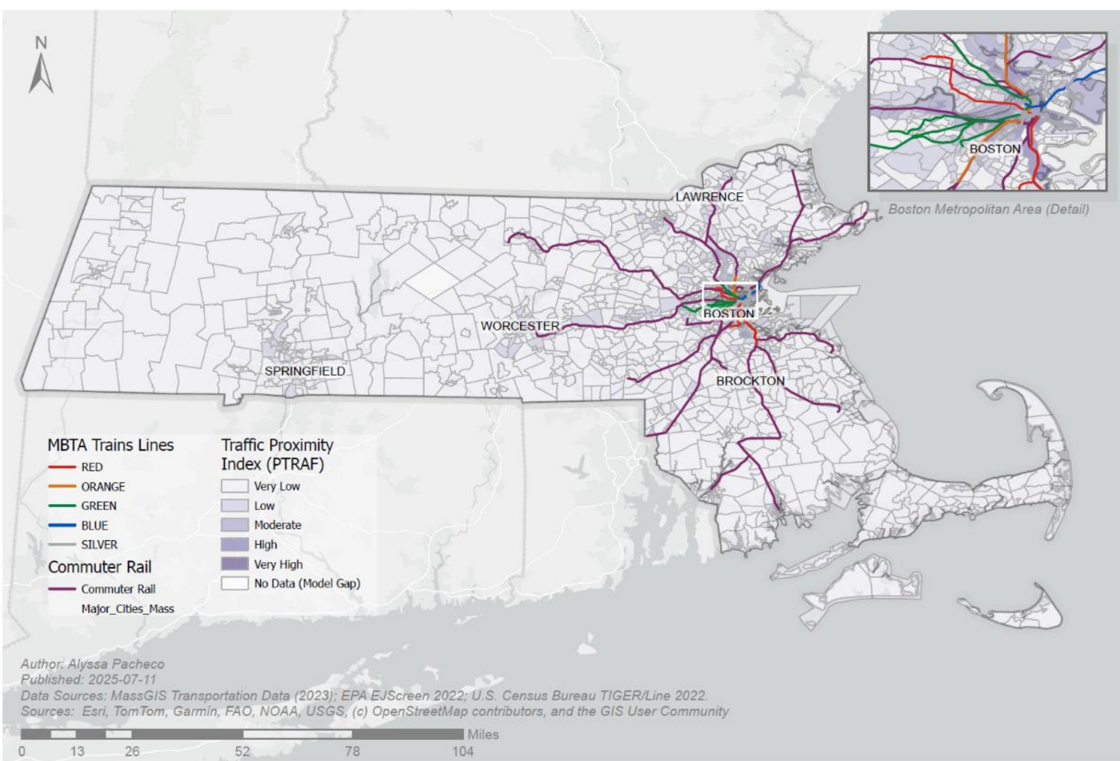


Traffic Proximity (PTRAF)

- PTRAF varied substantially across the state.
- **High-PTRAF tracts** clustered around:
    - Boston metro area
    - Worcester
    - Springfield
    - Brockton
    - Lawrence
- Patterns aligned with:
    - Dense transportation networks
    - Highways (I-93, I-95, I-90, I-290, Route 24)
    - MBTA commuter rail corridors

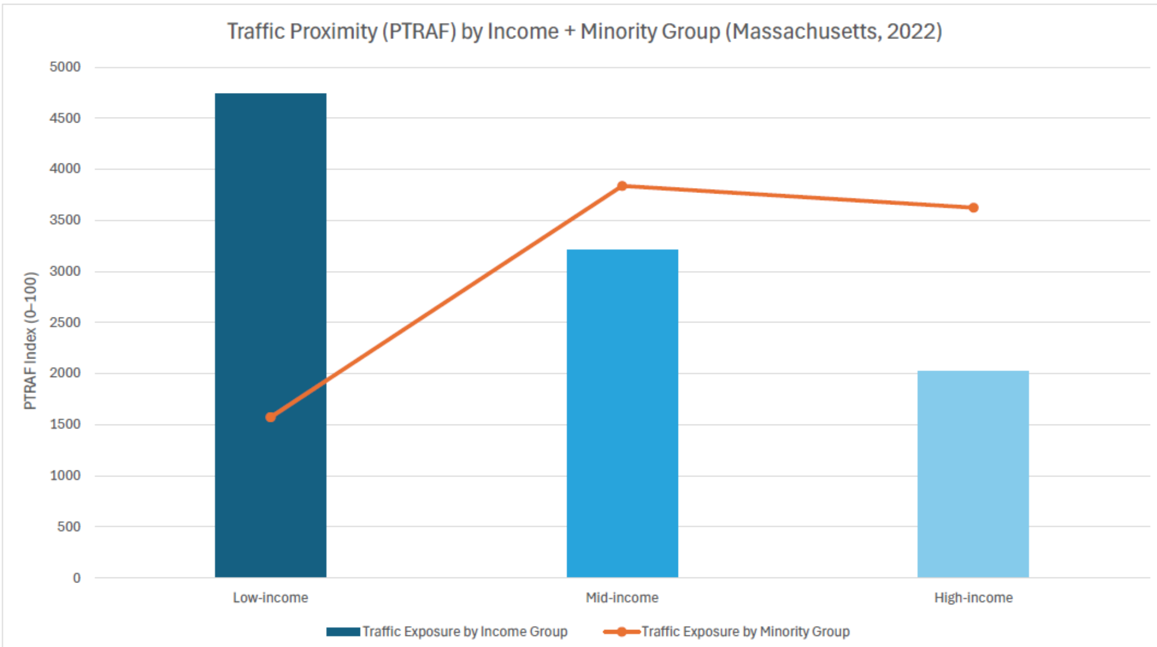
Traffic Proximity Across Massachusetts, 2022

Modeled exposure to high-traffic corridors by census tract (EPA EJScreen)



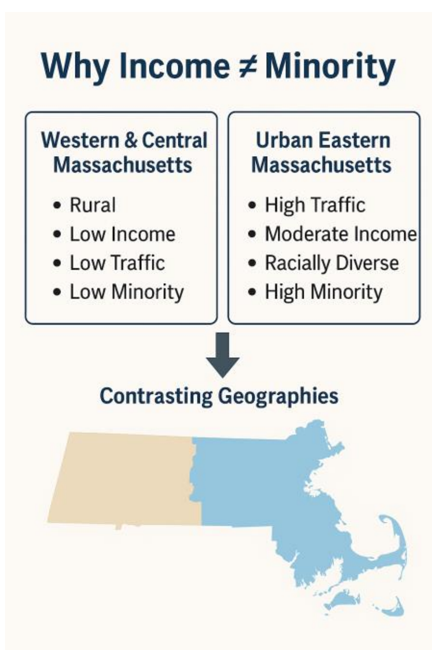
Group Differences

- **Low-Income Census Tracts** showed the **highest PTRAF exposure**.
- **High-Minority Tracts** exhibited **elevated PTRAF values** in metro areas.



However:

- Some **low-income tracts** in Western MA had **low minority percentages**, explaining the mismatch between income and minority trends.



Key Insights

1. **Ambient $PM_{2.5}$ is not an EJ issue in Massachusetts** — exposure is uniform statewide.
2. **Transportation exposure is an EJ issue** — disparities appear clearly in PTRAF.
3. **Low-income + high-minority tracts** in metro areas experience the **highest exposure**.
4. **Demographic geography matters:**
   - Western MA: low-income, low-minority, low PTRAF
   - Eastern MA: mid/high-income, high-minority, high PTRAF
5. **Transportation context explains the disparities** more than $PM_{2.5}$ alone can.
6. This analysis demonstrates the value of multi-layer GIS analysis when evaluating EJ risk.

Included Figures

✓ $PM_{2.5}$ Across Massachusetts (2022)
Statewide uniformity shown through continuous gradient.

✓ Traffic Proximity (PTRAF) Across Massachusetts (2022)
Includes MBTA commuter rail, state boundary, inset of Boston metro area, and labeled major cities.

✓ PTRAF Exposure by Income Group (Bar Chart)
Highlights how low-income tracts experience the highest proximity to major traffic sources.

✓ "Why Income ≠ Minority" Infographic
Explains how Massachusetts' demographic geography creates contrasting patterns.

Conclusion

This analysis reveals that while Massachusetts maintains relatively equitable ambient air quality, transportation-related exposures show meaningful disparities aligned with income, minority status, and urban transportation infrastructure. Integrating EJScreen indicators with transportation layers provides a deeper, more accurate understanding of environmental justice conditions in the state.

The findings demonstrate how GIS can uncover nuanced patterns not visible through a single indicator alone. This project serves as a foundation for future work, such as:

- Diesel PM (DSLPM) mapping
- Policy-focused transit exposure research
- Interactive dashboard and StoryMap development